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M. PHARM
(SEM II) THEORY EXAMINATION 2022-23
ADVANCED BIOPHARMACEUTICS AND PHARMACOKINETICS

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 2 = 20**

- (a) Define total clearance
- (b) Elaborate Bioequivalence.
- (c) Explain loading dose and maintenance dose.
- (d) Write down the Fick's law of diffusion.
- (e) Illustrate relative and absolute bioavailability.
- (f) Write down the Michael Menten equation.
- (g) Elaborate C_{max} , t_{max} and AUC.
- (h) Elaborate cross over design
- (i) Explain Gene therapy.
- (j) What is active transport?

SECTION B

2. Attempt any twoparts of the following:**2 x 10 = 20**

- (a) How a physicochemical nature of drug formulation affects drug product performance? Explain.
- (b) Discuss method of enhancement of bioavailability.
- (c) What do you understand by *in vitro-in vivo* correlation? Explain its significance.

SECTION C

3. Attempt any fiveparts of the following:**7 x 5 = 35**

- (a) Explain Pharmaceutical Dosage form factors affecting drug absorption in GIT.
- (b) Discuss various compendia methods of drug dissolution.
- (c) Derive an expression for calculating various pharmacokinetics parameters for a drug administered by IV bolus administration.
- (d) Illustrate steps involved in bioequivalence study and its clinical significance.
- (e) Elaborate different mechanisms of drug absorption through GIT.
- (f) Classify drug interactions. Discuss drug interaction linked to transporters.
- (g) Explain pharmacokinetics and pharmacodynamics characteristics essential for a drug in designing of controlled release products.